

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA1402

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO4: Examine intermediate code generation techniques to enhance code optimization (BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

CLASS TEST

Code of Conduct:

I VANDHANA VARSHA V(192524265) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: VANDHANA VARSHA V

Q. No	Understanding of Concepts (25)	Explanation (25)	Application (20)	Organization (15)	Accuracy(10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

Annexure A

CLASS TEST

Q1. Construct three address code for the following

if((a < b) and ((c < d) or (a > d))) then

z=x + y *z

else

z = z + 1

Q2.Translate (a < b) and (c < d) or (e < f) into three address statements using back patching.

Q3. Construct three address code using the following production for Boolean expression (a > b) or (c < d) or (e < f)

$B \rightarrow B1 \parallel B2$

$B \rightarrow B1 \ \&\& \ B2$

$B \rightarrow ! \ B1$

$B \rightarrow E1 \ \text{relop} \ E2$

$B \rightarrow \text{true}$

$B \rightarrow \text{false}$

Q4. Generate the three address code for the following program fragment-

while (A < C and B > D) do

if A = 1 then C = C + 1

else while A < = D do

A = A + B

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	15%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps

18/8/26

COMPILER DESIGN

Class test CO4-ATI

1) Construct Three address code:

if $(a < b)$ and $((c < d) \text{ or } (a > d))$ then

$Z = x + y * z$

else

$Z = Z + 1$

Ans given :

if $(a < b)$ and $((c < d) \text{ or } (a > d))$ then

$Z = x + y * z$

else

$Z = Z + 1$

Condition

$a < b$

$c < d$

$a > d$

L_1 : if $a < b$ goto L_2
goto L_5

L_2 : if $c < d$ goto L_4
goto L_3

L_3 :
if $(a > d)$ goto L_4
goto L_5

Three address code

$L_4 \Rightarrow Z = x + y * z$

L_4
 $t_1 = y * z$
 $t_2 = x + t_1$ goto L_6
 $Z = t_2$

$L_5 \Rightarrow Z = Z + 1$

$\therefore L_5$:

$t_3 = Z + 1$

$Z = t_3$

38/40

2) Translate $(a < b)$ and $(c < d)$ or $(e < f)$ into three address statement using Back patching

Ans $[(a < b) \text{ AND } (c < d)] \text{ or } (e < f)$

1) if $(a < b)$ goto 3

2) goto 5

3) if $(c < d)$ goto 7

4) goto 8

5) if $(e < f)$ goto 7

6) goto 8

7) True

8) False

3) Construct three address code using the following production for boolean expression $(a > b)$ or $(c > d)$ or $(e < f)$

$B \rightarrow B_1 \text{ || } B_2$

$B \rightarrow B_1 \text{ \&\& } B_2$

$B \rightarrow ! B_1$

$B \rightarrow E_1 \text{ relop } E_2$

$B \rightarrow \text{true}$

$B \rightarrow \text{false}$

Ans $L_1:$

if $(a > b)$ goto L_{TRUE}

goto L_2

L2:

if $C < D$ goto LTRUE
goto L4

L3:

if $E < F$ goto LTrue
goto LFalse

LTRUE

Boolean expression is true

LFALSE:

Boolean expression is False.

4) generate the three address code for the following program fragment

while $(A < C \text{ and } B > D)$ do

if $A = 1$ then $C = C + 1$

else while $A \leq D$ do

$A = A + B$

L1:

if $A < C$ goto L2
goto LEND

L2:

if $B > D$ goto L3
goto LEND

L3:

if $A == 1$ goto L4

goto L5

L4:

$t_1 = C + 1$

$C = t_1$

goto L1

L5:

if $A < D$ goto L6

goto L1

L6:

$t_2 = A + B$

$A = t_2$

